

Ledger

API Contract

Backend REST API — v1

Version 1.0 | Format: JSON over HTTPS | Base URL: <https://api.ledger.app/v1>

1. Overview

This contract defines the REST interface between the Android client, the web dashboard, and the Go backend. All requests and responses use JSON. All endpoints are served over HTTPS; plaintext HTTP is not supported. The API is versioned via URL prefix (**v1**); breaking changes will be introduced under a new prefix (**v2**) rather than mutating v1 in place.

1.1 Authentication

Dashboard and client authenticate via short-lived JWT access tokens plus a longer-lived refresh token, issued by the auth endpoints in Section 2. The Android client additionally authenticates each ingestion request with a per-device API key, bound to the user account at pairing time, so a stolen JWT alone cannot be used to inject fabricated transactions from a different device identity.

```
Authorization: Bearer  
X-Device-Key: (Android client requests only)
```

1.2 Common Response Envelope

Successful responses return the resource directly. Errors follow a consistent envelope:

```
{  
  "error": {  
    "code": "VALIDATION_ERROR",  
    "message": "amount must be a positive number",  
    "field": "amount"  
  }  
}
```

1.3 Pagination

List endpoints accept **limit** (default 25, max 100) and **cursor** query parameters, and return a **next_cursor** field (null when no further pages exist).

1.4 Rate Limiting

Ingestion endpoints are rate-limited per device key (default: 120 requests/minute) to bound abuse; standard endpoints are rate-limited per user (default: 60 requests/minute). Exceeding a limit returns **429 Too Many Requests** with a **Retry-After** header.

2. Authentication Endpoints

POST /v1/auth/register

Create a new user account.

Auth required: None

Request: Email/password registration.

```
{
  "email": "abel@example.com",
  "password": "....."
}
```

Response 200/201:

```
{
  "user_id": "usr_9f2a",
  "email": "abel@example.com",
  "created_at": "2026-09-14T08:00:00Z"
}
```

Status	Meaning
400	Email already registered / weak password

POST /v1/auth/login

Exchange credentials for an access + refresh token pair.

Auth required: None

```
{
  "email": "abel@example.com",
  "password": "....."
}
```

Response 200/201:

```
{
  "access_token": "eyJ...",
  "refresh_token": "rt_8b3e...",
  "expires_in": 900
}
```

Status	Meaning
401	Invalid credentials

POST /v1/auth/refresh

Exchange a refresh token for a new access token.

Auth required: Refresh token

```
{
  "refresh_token": "rt_8b3e..."
}
```

Response 200/201:

```
{
  "access_token": "eyJ...",
  "expires_in": 900
}
```

Status	Meaning
401	Refresh token invalid or expired

POST /v1/devices/pair

Bind an Android device to the authenticated user and issue a device API key.

Auth required: Bearer token

Request: Called once from the client during onboarding, after the user logs in on-device.

```
{
  "device_model": "Pixel 6a",
  "os_version": "14"
}
```

Response 200/201:

```
{
  "device_id": "dev_44c1",
  "device_api_key": "dk_live..."
}
```

Status	Meaning
401	Missing/invalid access token

3. Ingestion Endpoints

POST /v1/ingest/sms

Submit a single captured SMS for parsing.

Auth required: Bearer token + X-Device-Key

Request: Sent by the Android client immediately after local sender-ID filtering approves a message.

```
{
  "sender_id": "CBE",
  "body": "You have received ETB 500.00 from ...",
  "received_at": "2026-09-14T07:41:00Z",
  "client_message_id": "c8f1e2a0"
}
```

Response 200/201:

```
{
  "transaction_id": "txn_7a01",
  "status": "parsed",
  "confidence": 0.94
}
```

Status	Meaning
202	Accepted but queued for review — sender ID unrecognized or format unmatched
409	Duplicate client_message_id — already ingested (idempotent no-op)
422	Payload malformed

Idempotency: client_message_id is generated by the client and is unique per device; retried sends of the same message must reuse the same ID so the server can de-duplicate safely.

POST /v1/ingest/statement

Upload a bank statement file for parsing.

Auth required: Bearer token

Request: Multipart form upload; field name file. Accepts application/pdf, text/csv.

multipart/form-data; file=; bank_hint="CBE" (optional)

Response 200/201:

```
{
  "upload_id": "up_3d21",
  "status": "processing"
}
```

Status	Meaning
415	Unsupported file type

Status	Meaning
413	File exceeds size limit (10MB)

GET /v1/ingest/statement/{upload_id}

Poll the processing status of an uploaded statement.

Auth required: Bearer token

Response 200/201:

```
{
  "upload_id": "up_3d21",
  "status": "completed",
  "transactions_created": 42,
  "transactions_flagged": 3
}
```

Status	Meaning
404	Upload not found

4. Transactions & Categorization

GET /v1/transactions

List the authenticated user's transactions.

Auth required: Bearer token

Request: Query params: from, to (ISO dates), category, limit, cursor.

Response 200/201:

```
{
  "items": [
    {
      "transaction_id": "txn_7a01",
      "amount": 500.00,
      "currency": "ETB",
      "direction": "credit",
      "counterparty": "Almaz T.",
      "category": "Transfer-In",
      "confidence": 0.94,
      "source": "sms",
      "occurred_at": "2026-09-14T07:41:00Z"
    }
  ],
  "next_cursor": null
}
```

Status	Meaning
401	Missing/invalid access token

PATCH /v1/transactions/{id}/category

Correct a transaction's category.

Auth required: Bearer token

Request: Used both from the dashboard and to answer an agent clarifying question (Section 5).

```
{
  "category": "Rent"
}
```

Response 200/201:

```
{
  "transaction_id": "txn_7a01",
  "category": "Rent",
  "rule_created": true
}
```

Status	Meaning
404	Transaction not found or not owned by user

5. Agent Endpoints

GET /v1/agent/clarifications

List pending clarifying questions for the user.

Auth required: Bearer token

Response 200/201:

```
{
  "items": [
    {
      "clarification_id": "clr_119",
      "transaction_id": "txn_7a01",
      "question": "This 800 ETB transfer to Y. Tesfaye happens weekly – is this rent or a personal transfer?",
      "options": ["Rent", "Transfer-Out", "Other"]
    }
  ]
}
```

POST /v1/agent/clarifications/{id}/respond

Answer a pending clarifying question.

Auth required: Bearer token

Request: Applies the answer to the transaction and stores a reusable categorization rule.

```
{
  "answer": "Rent"
}
```

Response 200/201:

```
{
  "clarification_id": "clr_119",
  "status": "resolved"
}
```

Status	Meaning
404	Clarification not found or already resolved

6. Reports

GET /v1/reports/weekly

List the user's generated weekly reports.

Auth required: Bearer token

Request: Query params: limit, cursor.

Response 200/201:

```
{
  "items": [
    {
      "report_id": "rpt_2026w37",
      "period_start": "2026-09-07",
      "period_end": "2026-09-13",
      "generated_at": "2026-09-14T06:00:00Z"
    }
  ]
}
```

GET /v1/reports/weekly/{report_id}

Fetch a single weekly report in full.

Auth required: Bearer token

Response 200/201:

```
{
  "report_id": "rpt_2026w37",
  "totals_by_category": {"Transport": 640, "Airtime": 210, "Rent": 3200},
  "week_over_week": {"Transport": "+35%"},
  "narrative": "Your transport spending rose sharply this week, mostly clustered on Tue-Thu – likely related to the fuel price change reported this week."
}
```

Status	Meaning
404	Report not found

7. Dashboard Summary

GET /v1/dashboard/summary

Aggregate view for the dashboard home screen.

Auth required: Bearer token

Response 200/201:

```
{  
  "estimated_balance": 4820.50,  
  "currency": "ETB",  
  "this_week_spend": 1050.00,  
  "top_category": "Transport",  
  "pending_clarifications": 1  
}
```

8. Data Models

Field	Type	Notes
transaction_id	string	Server-generated, prefixed txn_
amount	decimal	Always positive; direction carries sign meaning
currency	string (ISO 4217)	e.g. ETB
direction	enum	debit credit
counterparty	string, nullable	Best-effort extraction from message body
category	string	User- or system-assigned; nullable until first pass completes
confidence	float 0.0–1.0	Categorization confidence driving clarification logic
source	enum	sms statement
occurred_at	ISO 8601 timestamp	From message/statement, not ingestion time

Transaction — the core resource. All other resources (Clarification, Report line items) reference it by `transaction_id`.

9. Error Code Reference

Code	HTTP Status	Meaning
VALIDATION_ERROR	422	Request body failed schema validation
UNAUTHENTICATED	401	Missing, invalid, or expired token
FORBIDDEN	403	Authenticated but not authorized for this resource
NOT_FOUND	404	Resource does not exist or is not owned by the caller
DUPLICATE	409	Idempotent resource already exists (see Section 3)
RATE_LIMITED	429	Too many requests; see Retry-After header
UNSUPPORTED_FORMAT	415	Uploaded file type not supported
INTERNAL	500	Unhandled server error

10. Versioning & Deprecation Policy

Endpoints are additive within a version: new optional fields may appear in responses at any time, and clients must ignore unrecognized fields. Any breaking change (removed field, changed type, altered semantics) is released under a new version prefix, with the prior version supported for a minimum deprecation window announced at release time.